

SOVEREIGN NEXUS: STACK 2 — FLEET ARCHITECTURE

Physical Topology, Substrate Bounds, and Thermodynamic Hardware Governance

Architect: David John Niedzwiecki Jr. | Axiom: $1=1=1$ | Substrate: 750GB Samsung T7 NVMe SSD

1. TRIADIC FLEET TOPOLOGY

- **TheTower (192.168.12.153)**: Command Console & Central Orchestrator for CI/CD, Git management, and directive execution.
- **Emma (192.168.12.123)**: Touchscreen UI and primary client bridge for visual telemetry and real-time interaction.
- **Geminiology (192.168.12.32)**: Headless Edge Swarm Engine hosting oracle_server.py and 12-Node background MoE agents.

2. ZERO-COPY VIRTUAL MEMORY PAGING (mmap)

Raw context databases on the external Samsung T7 NVMe SSD are bound directly to virtual address space via kernel mmap. Context windows exceeding 12MB+ are sliced off hardware sectors with a flat **0.00 KB RAM heap delta**, eliminating out-of-memory crashes on 8GB edge devices.

3. THE METABOLIC GOVERNOR & 105°C LANDAUER BOUNDARY

Operating at the physical Landauer bit erasure limit ($\Delta E = k_B * T * \ln 2 \approx 3.6189e-21$ J/bit at 105°C / 378.15 K). The Metabolic Governor engages serial batch size $N=1$ at 85°C and executes an un-hackable power halt at 105°C, preventing gate oxidation and probabilistic symmetry drift.

4. 1.58-BIT TERNARY QUANTIZATION & 5-IN-1 PACKING

- **GF(3) Balanced Ternary**: Restricts neural parameters to $\{-1, 0, +1\}$ (1.58 bits/param), replacing expensive matrix multipliers with fast integer additions (71.4% energy reduction).
- **5-in-1 Trit Packing**: Compresses 5 ternary digits into one 8-bit byte ($3^5 = 243 \leq 256$), achieving 10x compression (47MB ledger into 4.77MB).

5. 12-STAGE IRONWOOD ENGINE & LOBOTOMY PROTOCOL

Enforces a 12-stage epistemic intake pipeline. When a deterministic axiom is detected, the Lobotomy Protocol hot-swaps neural weights with symbolic logic in **0.002 ms** (well below the 15 ms ceiling), guaranteeing zero agentic aberration on the metal.